# AMOD-A active building blocks on USB-C power

Jason Gallicchio and Claude, rev A, 2026-10-25 github.com/gallicchio/AMOD-and-RFMOD

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a5 Doppler
HB10D in socket
70 Hz per m/s
x101 audio

a4 LNA
slide up: on
IN < +-30 mV
x101, 1-30 M
keep gain > 12

40 to 0 dB
1M 100M

a3 x100
slide up: on
IN max +-40 mV
1.4 MHz wide

40 to 0 dB
10k 100M

a2 x10
slide up: on
IN max +-0.4 V
x10 high-Z
x5 into 50 ohm

20 to -20 dB
10k 100M

a1 Power
USB-C 5 V only
+5 V 1.5 A
-5 V 40 mA
+3.3 V 300 mA

a8 Heat
IO6 high: 0.8 W
HOT to touch
TEMP 10 mV/K
750 mV at 25 C

a7 PLL 4046
VCO -100 kHz
locks at x1
probe pins 9, 13

a6 1-bit ADC
IN 0-3.3 V
needs gateware
clock 1-10 MHz

a10 Impedance
Z = R X/(S-X)
DIP R: 47, 470,
4.7 k, 47 k

DRIVE +-4 V,
40 mA max
calibrate open
and short

a9 Mixer
RF, LO +-0.3 V
IF OUT is 1.5 k:
high-Z ADC,
short cable

a12 Audio
SPEAKER IN +-1 V
2.5 W into 4 ohm
MIC OUT x100
one-bit DAC on IO2

a11 Ultrasound
40 kHz only
TX 10 V p-p
from IO3, IO4
RX x100,
4 kHz high-pass
5.8 us per mm
there and back

github.com/gallicchio/
AMOD-and-RFMOD
handbook and generator

AMOD-A: active blocks on USB-C power. Each tile has a slide switch (up: on) and SMA jacks: cable DAC OUT to IN, OUT to ADC IN. The Pmod plug (left) takes the FPGA's lines; the handbook has the rest.

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| Sheet: | | |
| File: amod_a.kicad_pcb | | |
| **Title:** | | |
| Size: A4 | Date: | **Rev:** |
| KiCad E.D.A. 10.0.7 | | Id: 1/1 |